
3AL Test 2 2022 Solutions

Time: 120 minutes

Available marks: 110

Maximum marks: 100

Instructions

1. Answer all questions in the lined answer booklets.
2. Start each question on a new page.
3. Make sure each answer is clearly labelled.
4. Show all work and reasoning.
5. This test consists of **5** questions.
6. If you've read the instructions write the name of your favourite mathematician on the front of your booklet (don't write it where *your* name is supposed to go) +1 mark

Good luck! :)

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1. (a) State the First Isomorphism Theorem.

First Isomorphism Theorem. *Let $\varphi : G \rightarrow G'$ be a group homomorphism. Then $\ker \varphi \triangleleft G$ and*

$$G/\ker \varphi \cong \operatorname{im} \varphi.$$

- (b) Let G be an abelian group. Fix a positive integer n . We define two subgroups:

$A = \{g \in G : g^n = 1\}$ and $B = \{g^n : g \in G\}$. Prove that $G/A \cong B$.

Consider the map $\varphi : G \rightarrow A$ defined by $\varphi(g) = g^n$ for each $g \in G$. For $g_1, g_2 \in G$, since G is abelian, we have

$$\varphi(g_1 g_2) = (g_1 g_2)^n = g_1^n g_2^n = \varphi(g_1) \varphi(g_2).$$

Thus φ is a homomorphism. Moreover, for each $a \in B$, we can write $a = g^n$, which follows from how B is defined. Then $\varphi(g) = g^n = a$ and so φ is an epimorphism. Moreover, $g \in \ker \varphi$ if and only if $\varphi(g) = 1$. That is, $g^n = 1$. Thus $\ker \varphi = A$. Then, by the First Isomorphism Theorem, $G/A \cong B$.

[5 + 10 = 15 marks]

2. (a) State the Orbit Decomposition Theorem.

Orbit Decomposition Theorem. *Let G be a group acting on a finite set X . Let $G * x_1, G * x_2, \dots, G * x_n$ denote the distinct nonsingleton orbits of X . Then,*

$$|X| = |X_f| + \sum_{i=1}^n |G : S(x_i)|.$$

- (b) Prove that if G is a p -group acting on a set X , then $|X| \equiv |X_f| \pmod{p}$.

Each index $|G : S(x_i)|$ in the Orbit Decomposition Theorem is a divisor of $|G|$. Thus each index must be some power of p . Each of these powers must exceed 0, since if $|G : S(x_i)| = p^0 = 1$ then $G * x_i$ is a singleton orbit, a contradiction. Thus p divides each $|G : S(x_i)|$. So p divides $|X| - |X_f|$. That is, $|X| \equiv |X_f| \pmod{p}$.

- (c) Let G be a group and H a normal subgroup of G .

- (i) Show that conjugation by G on H is a group action.

- (ii) Let p be a prime. Prove that if G is a p -group, then $H \cap Z(G)$ is nontrivial.

(i) Consider the map $*$: $G \times H \rightarrow H$ defined by $g * h = ghg^{-1}$. Since H is normal, $g * h$ is indeed an element of H and so $*$ is well-defined. Then, for $g_1, g_2 \in G$, we have

$$g_1 * (g_2 * h) = g_1 * (g_2 h g_2^{-1}) = g_1 g_2 h (g_1 g_2)^{-1} = g_1 g_2 * h.$$

Moreover, $1 * h = 1h1^{-1} = h$. Thus $*$ is a group action.

(ii) We consider the group action of conjugation by the p -group G on H . Since G is a p -group, we have $|H| \equiv |H_f| \pmod{p}$. Moreover, $|H|$ is a nontrivial subgroup of the p -group G , we must have $p \mid |H|$. Thus $0 \equiv |H_f| \pmod{p}$. However,

$$\begin{aligned} H_f &= \{h \in H : ghg^{-1} = h \text{ for all } g \in G\} \\ &= \{h \in H : gh = hg \text{ for all } g \in G\} \\ &= H \cap Z(G). \end{aligned}$$

Therefore, p divides $|H \cap Z(G)|$. Since H and $Z(G)$ are both subgroups of G , their intersection contains the identity so it is impossible for $|H \cap Z(G)| = 0$. Thus $1 < p \leq |H \cap Z(G)|$ and $H \cap Z(G) \neq \{1\}$.

[5 + 5 + 5 + 10 = 25 marks]

3. (a) State the definition of a normaliser.

Definition (Normaliser). Let G be a group and H a subgroup of G . The **normaliser** of H is

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

- (b) State the Second Sylow Theorem.

Second Sylow Theorem. Let G be a group of order $p^n m$ where $n \geq 1$ and p is a prime that does not divide m . Any two Sylow p -subgroups are conjugate.

- (c) Let G be a group of order $p^n m$ where $n \geq 1$ and p is a prime that does not divide m . Let H and K be Sylow p -subgroups of G .

(i) Prove that if $K = aHa^{-1}$ for some $a \in G$, then $H = a^{-1}Ka$.

(ii) Prove that $N_G(H)$ and $N_G(K)$ are conjugates.

(i) Let $h \in H$. Then $aha^{-1} \in K$ and so $h = a^{-1}ka$ for some $k \in K$. That is, $h \in a^{-1}Ka$, and so $H \subseteq a^{-1}Ka$. Now take $a^{-1}ka \in a^{-1}Ka$. Since $K = aHa^{-1}$, we can write $k = aha^{-1}$ for some $h \in H$. Thus $a^{-1}ka = h \in H$. That is, $a^{-1}Ka \subseteq H$.

(ii) Since H and K are Sylow p -subgroups, by the Second Sylow Theorem, we have $K = aHa^{-1}$ for some $a \in G$. We claim that $N_G(K) = aN_G(H)a^{-1}$. Take $g \in N_G(K)$. We need to show that $g \in aN_G(H)a^{-1}$. That is, $a^{-1}ga \in N_G(H)$. We have

$$a^{-1}gaH(a^{-1}ga)^{-1} = a^{-1}gaHa^{-1}g^{-1}a = a^{-1}gKg^{-1}a = a^{-1}Ka = H.$$

Thus $N_G(K) \subseteq aN_G(H)a^{-1}$. Now take $aga^{-1} \in aN_G(H)a^{-1}$. We need to show that this element belongs to $N_G(K)$. Notice that

$$aga^{-1}K(aga^{-1})^{-1} = aga^{-1}Kag^{-1}a^{-1} = agHg^{-1}a^{-1} = aHa^{-1} = K.$$

So, $N_G(K) = aN_G(H)a^{-1}$. Thus $N_G(K)$ and $N_G(H)$ are conjugates, as desired.

$[5 + 5 + 5 + 10 = 25 \text{ marks}]$

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4. (a) State the Third Sylow Theorem.

Third Sylow Theorem. *Let G be a group of order $p^n m$ where p is a prime that does not divide m . Let n_p denote the number of Sylow p -subgroups of G . Then $n_p \equiv 1 \pmod{p}$ and n_p divides m .*

- (b) Prove the Third Sylow Theorem, omitting the proof that $n_p \mid m$.

Let H be a Sylow p -subgroup of G and let X denote the set of all conjugates of H in G . By the Second Sylow Theorem, the number of Sylow p -subgroups of G equals the number of conjugates of H . That is, $n_p = |X|$.

Let H act on X by conjugation where $h * K = hKh^{-1}$. Since $n_p = |X|$, we have $n_p \equiv |X_f| \pmod{p}$. Note that $K \in X_f$ if and only if $h * K = hKh^{-1} = K$ for all $h \in H$. That is precisely $H \subseteq N_G(K)$. Then both H and K are subgroups of $N_G(K)$. Since they are Sylow p -subgroups of G , they must also be Sylow p -subgroups of $N_G(K)$. However, K is normal in $N_G(K)$, so it is the unique Sylow p -subgroup. Thus $H = K$. Therefore, $X_f = \{H\}$. Hence $n_p \equiv 1 \pmod{p}$, as desired.

[5 + 15 = 20 marks]

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5. (a) State the definition of a simple group and give an example.

Definition (Simple group). A group G is called **simple** if its only normal subgroups are the nonproper subgroup G and the trivial subgroup $\{1\}$.

- (b) Show that any group of order $520 = 2^3 \times 5 \times 13$ is not simple.

NOTE: The divisors of 104 are 1, 2, 4, 8, 13, 26, 52, 104. Moreover, $4 \times 26 = 104$ and $12 \times 40 = 480$.

By the Third Sylow Theorem, $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 104$. Given the divisors of 104, we see that the only options are $n_5 = 1$ or 26. If $n_5 = 1$, there is a unique Sylow 5-subgroup and it is normal and so G is not simple and we are done. So, assume that $n_5 = 26$. Similarly, $n_{13} \equiv 1 \pmod{13}$ and $n_{13} \mid 40$. So either $n_{13} = 1$ or 40. Again, if $n_{13} = 1$, we are done. So, assume $n_{13} = 40$.

We are assuming $n_5 = 26$ and $n_{13} = 40$. Our aim is to reach a contradiction, which will mean that either $n_5 = 1$ or $n_{13} = 1$, and so G fails to be simple.

Let $P_1, \dots, P_{26}$ be the Sylow 5-subgroups of G . By Lagrange's Theorem, the possible orders for the intersection of distinct Sylow 5-subgroups are 1 and 5. However, it cannot be 5 since the Sylow 5-subgroups are assumed to be distinct. Thus $P_i \cap P_j = \{1\}$ for $i \neq j$. Each Sylow 5-subgroup consists of 4 unique elements of order 5, not appearing in any other Sylow 5-subgroup. Thus there are $4 \times 26 = 104$ elements of G with order 5.

Through similar reasoning, there are $12 \times 40 = 480$ elements of G of order 13. This means there are at least $480 + 104 = 584$ elements in G . However, $|G| = 520 \leq 584$. So, it is impossible for both $n_5 \neq 1$ and $n_{13} \neq 1$ and we are done.

[5 + 20 = 25 marks]